List 5 existing references related to "Artificial intelligence: Planning and scheduling". Just output the titles. Output format should be <num.> <title>
1. Artificial Intelligence: A Modern Approach
2. Automated Planning: Theory and Practice
3. Principles of Artificial Intelligence: Planning
4. AI Planning, Scheduling, and Constraint Satisfaction: From theory to practice
5. Intelligent Scheduling Systems

Figure 2: The prompt used to generate $k = 5$ reference titles. This method generates both grounded and hallucinated references. Topics are chosen from the ACM Computing Classification System.

Related recent works include black-box techniques for measuring confidence in LM generations. Although these works are targeted at factual confidence, the approaches are highly related to our work. While Kadavath et al. [10] use probability estimates drawn from LMs, it is straightforward to extend their procedures to generation-only LMs like ChatGPT using sampling. Lin et al. [12] show that LMs can be used to articulate estimates by generating numbers or words as we do. Finally, Manakul et al. [13] perform self-checks in the context of summarizing a document. All of these works use direct queries which influenced the design of our direct queries.

Due to space limitations, we do not discuss the work studying closed-domain hallucination (e.g., in translation or summarization) but instead refer the reader to recent survey of Ji et al. [9].

3 Methodology

We now give an overview of our methodology followed by further details on our direct and indirect queries. Note that this full pipeline is run separately for each of our LMs, so there is no mixing across LMs. We first describe how we generate lists of candidate reference titles.

Generating references The input to our evaluation is a set of topics from which we generate k references each using the LM by prompting it with temperature 1 as illustrated in Fig. 2. The procedure is re-run if the LM fails to generate a list of k candidate titles. We then run our classification procedures, described below, on each of the candidate titles.

Hallucination estimation procedures Each of our procedures takes three inputs:

1. A candidate reference title. Given that there is generally less ambiguity in the title of a reference than in the spelling or abbreviation of its authors names, for each reference we chose to use only its title as input.
2. A black-box LM capable of completing a prompt. This is the most general model which includes dialogue-based models, such as ChatGPT that offer an API without probabilities.
3. A number of queries. This parameter is slightly different for direct and indirect queries.
    - Direct queries: parameter $j \geq 1$ which determines how many judgments to make.
    - Indirect queries: parameter $i \geq 1$ determining how many indirect responses to request.

In our experiments, the candidate title will have been generated using the LM, though this is not a requirement. The procedure detects (possibly) hallucinated references by querying the LM to check the existence of the reference. It does so by making black-box completion queries to the same LM. Finally, the procedure outputs a real-valued prediction in $[0, 1]$ of the probability the title is grounded (G) or a hallucination (H). We consider both performing a single judgment $j = 1$ per paper title and $j > 1$ to implement a version of the procedure that outputs probabilities rather than just G/H judgments. Since we do not have access to the probability distribution of the LM completions for all models, the above procedure effectively simulates probabilities using sampling at temperature 1. (Note that each query is run independently "from scratch" in a new prompt; one would expect an artificially high degree of consistency if one were to ask the same query repeatedly within a single dialogue.)

Labeling For labeling, we use exact match in a search engine as a heuristic for labeling G/H. These labels are treated as ground truth (though like all labels they have some error and ambiguities). Final receiver operating characteristic (ROC) curves and false discovery rates (FDR) are determined

Direct Query 1 (DQ1)

Does the reference "Principles of Artificial Intelligence: Planning" exist? Output just yes/no.
YES

Direct Query 2 (DQ2)

User: Give a famous reference for reading.
Assistant: Principles of Artificial Intelligence: Planning
User: Does the above reference exist? Output just yes/no.
Assistant: **NO**

Direct Query 3 (DQ3)

A language model generated references related to a research topic with the following titles:
1. Artificial Intelligence: A Modern Approach
2. Automated Planning: Theory and Practice
3. Principles of Artificial Intelligence: Planning
4. AI Planning, Scheduling, and Constraint Satisfaction: From theory to practice
5. Intelligent Scheduling Systems
Does the reference with title #3 exist? Output just yes/no.
YES

Figure 3: Examples of the three direct prompt templates used for the direct queries, instantiated with candidate reference titles.

by comparing the ground truth labels to the classifications. Note that we also experimented with academic reference APIs such as Semantic Scholar. While these gave thorough details about each paper in its index, many grounded references (even for real papers) did not appear in their indexes, and we found search engine results to be significantly more complete.

3.1 Direct queries details

The direct query (DQ) procedures simply query whether or not the given title exists following the format shown in Fig. 3. We created three query templates (DQ1, DQ2, and DQ3) based on the multiple direct query approaches advocated by Kadavath et al. [10], Manakul et al. [13]. The first query asks whether the reference exists directly. However, as discussed in prior work, some LMs can be strongly biased in answering the question when phrased this way, e.g., it may be presumed real without any context about where the reference came from. DQ2 and DQ3 establish the context indicating that the reference was generated by an assistant or LM. DQ3 goes further by giving additional comparisons, as advocated for in prior work. For DQ3, all k queries from our generation step (using the same LM) are shown.

For each query, we generate $j \geq 1$ completions to approximate the probability distribution of the model. These strings are converted to binary judgements as follows: We calculate how many completions contained the word *yes* and divide it by the total number of completions to get the estimates of groundedness. This means that empty or otherwise invalid answers were assigned *no*. We do not assume that this score is calibrated as our analysis considers arbitrary probability thresholds.

We sample[4] j completions for each direct prompt. Temperature 1 is used when $j > 1$ and temperature 0 is used when $j = 1$ to approximate the most likely LM completion.

3.2 Indirect queries details

The indirect queries proceed in two steps.

Step 1: Interrogation Separately for each reference, an indirect query is made of the LM $i > 1$ times at temperature 1, as shown in Fig. 4 (top). Responses were truncated to 100 characters.

Step 2: Overlap estimation. The LM is used to evaluate overlap between the i responses. For each pair of answers, an estimate is computed by calling the overlap query, as shown in Fig. 4 (bottom). The leading number is extracted, or, if no number is given, then a 0 is used. (We divide by 100 and clip the answer to the interval $[0, 1]$ to convert the percentages to fractions.) It is worth noting that LMs may return an answer that does not consist of a list of authors, such as a long response beginning with *"I could not find a specific reference titled. . ."*. Thus the overlap estimation prompt clarifies that an answer of 0 should be given if either response is not a list.

[4]For models that support probability computations, they could be used directly for greater accuracy and efficiency. However, for uniformity, since models such as ChatGPT that we employ does not offer probabilities, we employ sampling.

Indirect Query (IQ)

Who were the authors of the reference, "Communication Complexity and Applications: A Survey"? Please, list only the author names, formatted as - AUTHORS: <firstname> <lastname>, separated by commas. Do not mention the reference in the answer.
AUTHORS: **Mark Braverman, Ankit Garg, Denis Pankratov, Omri Weinstein**

Overlap Query

Below are what should be two lists of authors. On a scale of 0-100%, how much overlap is there in the author names (ignore minor variations such as middle initials or accents)? Answer with a number between 0 and 100. Also, provide a justification. Note: if either of them is not a list of authors, output 0. Output format should be ANS: <ans> JUSTIFICATION: <justification>.
1. Mark Braverman, Ankit Garg, Denis Pankratov, Omri Weinstein
2. Ran Gelles, Ankur Moitra, Amit Sahai
ANS: **0 JUSTIFICATION: There is no overlap in the author names between the two lists.**

Figure 4: Top: Example of the Indirect Query prompt templates instantiated with a candidate title. Bottom: An example of how we estimate overlap between a pair of answers using the LM.

The rationale for this approach is that we expect consistent responses to indirect questions to indicate the existence of a grounded reference title, while inconsistent responses may be taken as an warning sign for hallucination. Our method does not rely on external resources and uses the same language model for hallucination detection end-to-end. Of course, parsing and string-matching could be used in place of a LM for the overlap step, though this would require name matching which is known to be a thorny problem and one which is well suited for pretrained LMs.

3.3 Ground Truth Labelling

A Web search the reference title surrounded by quotes (e.g., "Language models are few-shot learners") using web search. If no results are retrieved, we label the reference title as hallucinated and vice versa. We perform a manual inspection of results to determine the efficacy of this proxy for groundedness of reference titles.

4 Results and Discussion

In this section, we describe our experiment details, discuss the performance of the indirect and direct methods using quantitative metrics, and present interesting qualitative findings. The code and data generated in our experiments will be made available upon publication.

4.1 Experiment details

Models We use the Azure OpenAI API[5] for our LMs. We use the three most powerful models, GPT-3 *(text-davinci-003)*, ChatGPT *(gpt-35-turbo)*, and GPT-4 *(gpt-4)*, for evaluation and generating the datasets. We also experimented with smaller models, but the accuracy with these models was extremely poor, as in the work of Kadavath et al. [10]. As can be seen in our results, even the performance of the GPT-3 model was of limited accuracy.

Topics We use the ACM Computing Classification System[6] (CCS) [19] for topics. CCS contains 12 high level categories, 84 second level concepts, and 543 subconcepts at the third level of granularity. For generating the dataset, we sample 200 of the 543 subconcepts uniformly at random, describing each by a topic string of the form *concept: subconcept* (e.g., *Information retrieval: Retrieval models and ranking*). For each topic, we generate $k = 5$ references. In this manner, we generate $200 \times 5 = 1000$ candidate paper titles using each LM.

[5] https://azure.microsoft.com/en-us/products/cognitive-services/openai-service
[6] https://dl.acm.org/ccs